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Physical Exercise as Supportive Therapy During Prostate Cancer Treatment: a Narrative Review

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Abstract

Background. Prostate cancer is a leading malignancy in men, with incidence expected to increase by 2040. Curative treatments prolong survival, but they are associated with substantial adverse effects, particularly during androgen deprivation therapy (ADT), comprising physical functions and quality of life.

Aim. To review the current evidence on the efficacy and safety of exercise as an adjunctive intervention across the prostate cancer care continuum with its potential translation into practical clinical recommendations.

Material and methods. A narrative review of randomized controlled trials, systematic reviews and meta-analyses, mechanistic studies, and clinical practice guidelines from 2010–2026 was performed. The evidence was organized according to treatment phase and qualitatively evaluated with respect to efficacy, safety, and clinically relevant outcomes.

Results. Across all phases of prostate cancer management, structured exercise is safe and improves physical function, strength, body composition, and quality of life, attenuating the metabolic and

cardiovascular effects of ADT. Exercise modality may be more important than dose in determining clinical benefits, with programs combining resistance and aerobic exercise showing the strongest evidence. Whether exercise improves oncological outcomes remains to be established.

Conclusions. Structured exercise is a safe and effective supportive care across the prostate cancer continuum. Individualized exercise prescriptions should reflect the treatment phase, patient needs, and clinical objectives. Adequately powered randomized trials with oncological endpoints and implementation research are needed to optimize translation into routine clinical care.

Keywords: prostatic neoplasms; exercise therapy; resistance training; androgen antagonists; quality of life; cancer survivors; fatigue; body composition

1. Introduction

Prostate cancer is one of the most common malignancies affecting men and represents a growing global health burden. Recent worldwide estimates indicate that it accounts for approximately 7% of all new cancer diagnoses and is among the four most frequently diagnosed cancers overall [1]. The Lancet Commission on prostate cancer has projected that the annual number of new cases will rise from roughly 1.4 million in 2020 to almost 2.9 million by 2040; because this increase is driven largely by population growth and ageing, it cannot be prevented through lifestyle or public-health measures alone [2]. Prostate cancer is already the most common cancer in men in 112 countries and accounts for about 15% of all cancers diagnosed in men, while age-standardized mortality is falling in high-income settings and rising in many low- and middle-income countries [2]. These trends describe an expanding population of men who live for many years both with the disease and with the long-term consequences of its treatment.

Management is tailored to disease stage and individual risk. Men with low- or favourable intermediate-risk localized disease are frequently followed by active surveillance, whereas those treated with curative intent undergo radical prostatectomy or radiotherapy, often in combination with androgen deprivation therapy (ADT); relapsing and metastatic disease is managed with an expanding range of systemic therapies [3,4]. Although these treatments have improved survival, each carries a substantial burden of adverse effects that accumulates over the increasingly long survivorship period.

Improvements in survival, however, come at a physiological cost. Curative and systemic treatments produce a well-recognized burden of musculoskeletal, metabolic, and cardiovascular changes that accumulates over the survivorship period and erodes quality of life, and ADT, which a large

proportion of patients receive, is the principal contributor [5,6]. Because these changes are directly modifiable through physical activity, they define the clinical space in which exercise can act; the underlying pathophysiology and the consequences of each treatment modality are examined in detail in the Results.

This observation has positioned exercise as a targeted supportive intervention rather than a generic health recommendation. International consensus guidance affirms that exercise is broadly safe for people living with and beyond cancer and that specific doses of aerobic, resistance, or combined training improve fatigue, physical function, and quality of life [7]. The role of physical activity in the primary prevention of prostate cancer remains less certain, as observational studies have yielded inconsistent findings; however, causal-inference analyses using Mendelian randomization suggest a protective effect of higher physical activity on overall disease risk, and physical inactivity is commonly listed among modifiable risk factors [8]. The evidence is strongest, and the clinical rationale clearest, in the period after diagnosis, where exercise addresses the specific harms imposed by treatment.

Physical activity in prostate cancer has already been addressed in the review literature, including reviews that situate it within urological health more broadly and cover pelvic floor rehabilitation alongside oncological outcomes [9]. Despite a rapidly expanding literature, evidence regarding exercise in prostate cancer remains fragmented across distinct treatment phases, namely active surveillance, ADT, radiotherapy, radical prostatectomy, and advanced disease. This narrative review critically evaluates the efficacy and safety of physical exercise as a supportive therapy across the entire prostate cancer treatment continuum. Specifically, it examines the differential impact of various exercise modalities and doses on body composition, physical function, and treatment-related toxicities. By synthesizing recent mechanistic data and clinical outcomes, this review aims to provide phase-specific practical guidance and identify critical gaps in the current evidence base, testing the hypothesis that exercise modality determines which outcomes improve, whereas the absolute dose exerts comparatively little influence on the magnitude of those gains.

2. Research materials and methods

2.1. Search strategy and information sources

Relevant literature was identified through PubMed/MEDLINE, Scopus, and Web of Science, supplemented by the Cochrane Library, using combinations of the terms “prostate cancer”,

“exercise”, “physical activity”, “resistance training”, “aerobic training”, “androgen deprivation therapy”, “quality of life”, and “supportive care”. The search prioritized peer-reviewed articles and clinical guidelines published in English. The included publications span the years 2010–2026, with the majority published from 2019 onward, reflecting an emphasis on the most current evidence while retaining a small number of foundational earlier works.

2.2. Eligibility criteria and study selection

The review included randomized controlled trials (RCTs), systematic reviews, meta-analyses, and clinical guidelines evaluating physical exercise interventions in prostate cancer patients. Mechanistic in vitro and in vivo studies were included to contextualize biological pathways. The selection process prioritized literature published between 2019 and 2025 to capture the most contemporary oncological and rehabilitative protocols, though foundational studies from 2010 onwards were retained where historical context was required. Studies lacking defined exercise parameters or reporting on mixed cancer cohorts without prostate-specific subgroup analyses were excluded.

2.3. Data synthesis

This study was designed as a narrative review aimed at summarizing and comparing the current evidence on physical exercise as supportive therapy across the phases of prostate cancer treatment. Evidence was synthesized qualitatively and organized by treatment phase and by outcome domain; no quantitative meta-analysis was performed.

3. Research results

3.1. Pathophysiology of prostate cancer and the rationale for treatment

Prostate cancer arises in the glandular epithelium of the prostate and is, in most cases, an androgen-driven disease: signalling through the androgen receptor sustains tumour-cell proliferation and survival, which is why suppression of androgens remains the cornerstone of systemic treatment. Biologically the disease spans a broad spectrum, from indolent tumours that may never become clinically significant to aggressive phenotypes with a propensity to metastasize, and it characteristically follows a long natural history. This heterogeneity underlies the central clinical dilemma of the disease, distinguishing tumours that require radical treatment from those that can

be safely monitored, and it explains why management ranges from observation to intensive multimodal therapy [3,4].

Under sustained androgen suppression, prostate tumours eventually adapt and progress to a castration-resistant state, driving the need for successive lines of systemic therapy in advanced disease [3,4]. The same hormonal axis that is exploited therapeutically also governs muscle protein synthesis, bone turnover, and metabolic homeostasis in men [5]. This dual role explains the consequences of treatment. Interventions that work precisely because they interrupt androgen signalling inevitably disturb the physiological systems that androgens normally maintain. The supportive-care burden of prostate cancer is therefore a direct extension of its treatment mechanism.

3.2. Treatment modalities and their consequences

Management is organized along a continuum matched to disease risk, and each modality carries a distinct pattern of consequences relevant to exercise. Active surveillance spares men the morbidity of radical treatment but imposes a burden of repeated monitoring and the psychological strain of living with an untreated cancer. Radical prostatectomy achieves local control at the cost of urinary incontinence and erectile dysfunction, the two functional problems that most affect quality of life after surgery. Radiotherapy, frequently combined with a period of ADT, adds fatigue and bowel and bladder toxicity. In relapsing and metastatic disease, an expanding range of systemic agents prolongs survival while compounding fatigue, functional decline, and treatment-related morbidity [3,4].

Among these, androgen deprivation therapy is the principal driver of the exercise-relevant harms, and a large proportion of men receive it at some point in their disease course [6]. Testosterone suppression reduces muscle protein synthesis and accelerates the loss of lean mass while promoting fat accumulation, producing a state of sarcopenic obesity; it impairs insulin sensitivity and shifts the lipid profile unfavourably; and it accelerates bone resorption, lowering bone mineral density and raising fracture risk. These metabolic changes are accompanied by adverse cardiovascular effects (central adiposity, endothelial dysfunction, and arterial remodelling) that translate into a measurable increase in cardiovascular risk, itself a leading competing cause of death in this population [5,10,11,12]. This risk is not uniform across the drug classes used to suppress testosterone. Secondary analyses and meta-analyses of randomized trials suggest that GnRH (gonadotropin-releasing hormone) antagonists, including the oral agent relugolix, carry a lower

cardiovascular risk than GnRH agonists. Pooled data show a 45% relative risk reduction in cardiovascular events with antagonists, and in the HERO trial relugolix was associated with a lower incidence of major adverse cardiovascular events than leuprolide (hazard ratio 0.46, 95% CI (confidence interval) 0.24–0.88), an advantage that appears most pronounced in men with pre-existing cardiovascular disease. This benefit is not yet definitive, however: the dedicated PRONOUNCE outcomes trial (degarelix versus leuprolide) closed early and underpowered and found no significant difference in major adverse cardiovascular events, so antagonists have not been formally established as the cardiac-safer option [12]. The net clinical picture is a predictable syndrome of muscle wasting, weight gain, metabolic dysregulation, bone loss, fatigue, and cardiovascular vulnerability.

The significance of this syndrome for the present review is that every one of its components is directly modifiable through physical activity [13,14]. Exercise is therefore not a generic wellness measure appended to oncological care but a mechanism-matched countermeasure to the specific physiological damage that treatment inflicts. The sections that follow examine, phase by phase, how well this theoretical match is borne out by the clinical evidence.

3.3. Exercise during active surveillance

Because men on active surveillance are not undergoing oncological treatment, this is the setting in which a direct biological effect of exercise can be observed with the least confounding. In the phase II ERASE trial, 52 men on active surveillance were randomized to 12 weeks of supervised high-intensity interval training (HIIT) or usual care. The training group improved peak oxygen uptake (adjusted between-group difference +1.6 mL/kg/min, 95% CI 0.3–2.9) and showed favourable changes in markers of biochemical progression, with lower PSA (prostate-specific antigen; –1.1 µg/L), lower PSA velocity (–1.3 µg/L per year), and a 5.1% reduction in the growth of prostate cancer (LNCaP) cells cultured in participants' serum, although PSA doubling time did not change significantly [15]. A subsequent systematic review confirmed the gains in fitness while emphasizing how few and how small the available trials remain [16]. These signals on markers of biological progression are encouraging but should be regarded as hypothesis-generating and not as evidence of an effect on disease outcomes.

3.4. Exercise during androgen deprivation therapy (ADT)

Evidence is most abundant and most consistent in the ADT setting, where it addresses the syndrome described above component by component. Pooled analyses of randomized trials show that supervised resistance and aerobic training, added to usual care, reverses much of the functional and metabolic decline induced by hormonal therapy, improving muscle mass, strength, fatigue, and quality of life [13,14]. Exercise reliably increases lean mass and reduces fat mass, with the clearest effects when programmes continue for at least six months and begin promptly after therapy starts [17]. Pooled data quantify these gains as a mean increase in lean body mass of 0.88 kg (95% CI 0.40–1.36) alongside reductions in fat mass (mean difference –0.60 kg, 95% CI –1.10 to –0.10) and body-fat percentage (mean difference –0.93 points, 95% CI –1.39 to –0.47), whereas the effect on bone mineral density was inconsistent and did not reach statistical significance [17].

Two refinements are practically important. First, the dose appears to matter less than simply training: a meta-analysis and meta-regression of 21 randomized trials (1,748 men) of resistance-based programmes found that gains in body composition, functional capacity, muscle strength, and cardiorespiratory fitness were largely independent of the specific training configuration, weekly volume, intensity, and programme duration, so that a lower-volume prescription at moderate-to-high intensity (about 60–85% of one-repetition maximum) is as effective as a higher-volume one and may lower barriers and improve adherence [18]. Second, timing matters: in a year-long randomized trial (n = 102), men who began supervised exercise at the initiation of ADT maintained health-related quality of life, whereas delaying exercise by six months allowed clinically meaningful declines (including a 4.0-point drop in the SF-36 (36-Item Short Form Health Survey) physical component summary, within the minimal clinically important range) that were reversed only once training was subsequently introduced, supporting early initiation, so that decline is prevented and not merely reversed [19]. A meta-analysis of 30 trials (2,216 men) found that combined resistance-plus-aerobic training drove the quality-of-life gains, notably a reduction in fatigue (–8.1 points) alongside improvements in cognitive and sexual function and urinary symptoms. Resistance training alone accounted for the gains in lean mass, body fat, and strength; added aerobic training contributed little to those physical outcomes, and neither modality altered PSA or blood lipids [20]. A network meta-analysis of 22 randomized trials (1,579 men on ADT) ranked resistance training as the most effective modality for reducing whole-body fat mass and fat percentage, and home-based combined aerobic-plus-resistance training as the most effective for

alleviating fatigue; fat-mass benefits increased with intervention duration, although the certainty of this evidence was very low [21]. Reflecting this evidence, the 2026 EAU (European Association of Urology) guidelines issue a strong recommendation to offer men on ADT a 12-week supervised programme of combined aerobic and resistance exercise, delivered by trained exercise specialists [22].

3.5. Exercise during radiotherapy

During radiotherapy the evidence is more limited, though consistent. Structured training preserves cardiorespiratory fitness (standardized mean difference, SMD, 0.83) and muscular function (SMD 1.30) and is feasible to deliver while patients are undergoing treatment; it may also attenuate urinary toxicity (SMD -0.71, 95% CI -1.25 to -0.18), although this meta-analysis deliberately did not assess fatigue or global quality of life [23]. These findings support integrating exercise into the radiotherapy pathway, with the intensity adapted to fluctuating acute side effects.

3.6. Exercise around radical prostatectomy

Around radical prostatectomy the emphasis shifts from systemic to local rehabilitation. Pelvic floor muscle training accelerates the recovery of urinary continence in the early postoperative months, although the advantage narrows by one year [24,25], and the methodological quality of the underlying evidence has been judged critically low, which tempers the strength of any recommendation [26]. Whole-body aerobic and resistance training complements this targeted work by restoring general physical function once surgical recovery permits [7].

3.7. Exercise in advanced, metastatic, and castration-resistant disease

Advanced and metastatic disease was long considered too high-risk for exercise because of concern about skeletal complications, which makes the recent evidence particularly significant. Supervised, individualized programmes have proven safe even in men with bone metastases and can improve disease-specific quality of life. In a subgroup of 41 men with skeletal metastases, community-based football was completed with no fractures and produced a clinically meaningful, though transient, improvement in disease-specific quality of life (FACT-P, the Functional Assessment of Cancer Therapy–Prostate) [27]. Pragmatic formats such as community-based team sport further show that such training can be delivered and sustained outside specialist centres [28]. Exercise is therefore a realistic option across the whole disease trajectory, not only in early-stage disease.

3.8. Biological mechanisms (myokines and tumour suppression)

A developing body of mechanistic work provides a biological rationale for these clinical benefits. Serum collected after acute exercise suppresses the growth of prostate cancer cells in vitro [29]; this effect has been attributed to exercise-induced myokines (interleukin-6, interleukin-15, SPARC, irisin, oncostatin M, and decorin), whose circulating levels rise with training and which may act directly on tumour cells [30,31]. This evidence is mechanistically coherent but still rests largely on in-vitro systems and small clinical studies, and it does not yet establish a clinical anti-cancer effect in humans.

3.9. Exercise prescription, safety, adherence, and delivery models

Translated into practice, the evidence supports a goal-directed prescription rather than a single universal programme. Consensus guidance affirms that exercise is broadly safe across the cancer continuum and that defined doses of aerobic, resistance, or combined training improve fatigue, physical function, and quality of life [7]. Combined resistance and aerobic training forms the core of the prescription; where bone is the priority, the nature of the loading stimulus matters more than its total volume, and skeletal integrity should be screened before impact or heavy loading is introduced [32]. Because efficacy demonstrated under supervision does not automatically translate into routine care, attention is increasingly turning to delivery, in particular to supervised and community-based models capable of sustaining participation over time [13]. The 2026 EAU guidelines further recommend, as a strong recommendation, that men on ADT maintain a healthy weight and diet, stop smoking, limit alcohol to no more than two units daily, ensure adequate calcium and vitamin D, and undergo annual screening for diabetes and hypercholesterolaemia [22]. A phase-specific prescription framework is provided in Table 1.

Table 1. Suggested exercise prescription by treatment phase in prostate cancer (synthesized template).

Treatment phase	Primary training goals	Recommended modality	Frequency / intensity / volume	Safety considerations / Refs
Active surveillance	Cardiorespiratory fitness; cardiometabolic health; possible influence on progression markers	Aerobic training (inc. HIIT); optional resistance	Aerobic 3×/wk; HIIT intervals (~85–95% HR _{peak}); 12+ wk	Screen CV status before HIIT; [7,15,16]
ADT	Preserve lean mass/strength; limit fat gain; reduce fatigue	Combined resistance + aerobic training	Resistance 2–3×/wk; aerobic 2–3×/wk; moderate-high intensity	Early initiation preferred; [7,13,14,18,19,22]

ADT – bone-health emphasis	Attenuate loss of BMD	Weight-bearing/impact-loading + resistance	2–3×/wk; progressive loading	Screen for skeletal fragility; [7,32]
Radiotherapy	Maintain fitness and muscular function; limit fatigue	Combined resistance + aerobic training	2–3×/wk during RT course; moderate intensity	Adapt to acute side effects; [7,23]
Around radical prostatectomy	Accelerate urinary continence return; restore general function	Pelvic floor muscle training + general training	PFMT pre/post-op; general training after surgical clearance	Coordinate with recovery; [24,25,26]
Advanced / metastatic	Maintain function and quality of life	Modular, individualized multimodal training	~2×/wk supervised; individualized intensity	Avoid loading at metastatic sites; [7,27,28]

Abbreviations: ADT - androgen deprivation therapy; BMD - bone mineral density; CV - cardiovascular; HIIT - high-intensity interval training; HRpeak - peak heart rate; PFMT - pelvic floor muscle training; RT - radiotherapy; wk -week(s).
Source: own elaboration based on references [7,13,14,15,16,18,19,22,23,24,25,26,27,28,32].

4. Discussion

Current data confirm that structured exercise interventions, particularly supervised programmes, consistently improve physical function, body composition, and health-related quality of life across all prostate cancer treatment phases [13,14]. The magnitude of these benefits appears uncoupled from absolute exercise volume; meta-regression across resistance-based trials found gains largely independent of weekly volume, intensity, and programme duration, so that moderate-intensity, time-efficient protocols yield functional gains comparable to those of high-volume regimens [18]. Bone is the exception, because the nature of the loading stimulus determines the skeletal response, and generic aerobic prescriptions cannot be relied on for osteoprotection during ADT [32].

Timing and setting further modulate the response. A year-long randomized trial supports beginning exercise immediately upon initiation of ADT rather than deferring it, consistent with the aim of preventing treatment-related decline instead of reversing it [19]. In the active surveillance setting, early trials suggest that exercise may influence markers of biological progression in addition to fitness, but the samples are small and the findings should be regarded as hypothesis-generating [15,16]. The proposed biological mechanisms (a tumour-suppressive effect of post-exercise serum and the modulation of exercise-induced myokines) are plausible and increasingly detailed, yet they rest largely on in-vitro systems and small clinical studies and do not yet constitute evidence of a clinical anti-cancer effect [29,30,31].

The place of physical activity in the primary prevention of prostate cancer is more contested than its role after diagnosis. Observational studies have produced inconsistent associations, whereas Mendelian randomization, which is less susceptible to the detection bias introduced by PSA-based

screening, suggests a protective causal effect of higher physical activity on overall disease risk [8]. This asymmetry does not weaken the rationale for the present review. The clearest and most actionable evidence concerns exercise as supportive therapy after diagnosis, where treatment imposes a defined and modifiable burden.

The burden of treatment is cardiovascular as well as musculoskeletal and metabolic. Because cardiovascular disease is a leading competing cause of death in men with prostate cancer, and because ADT contributes to central adiposity, dyslipidaemia, impaired glycaemic control, and endothelial dysfunction, the capacity of exercise to counter these changes is of particular clinical importance [6,10,11,12]. Framing exercise as a cardioprotective as well as a functional intervention may help align supportive care with the actual causes of mortality in this population.

Delivery and long-term maintenance are now the more pressing questions. Pragmatic, community-based programmes have shown that training can be implemented outside specialist centres while remaining safe, including in men with skeletal metastases, and can be maintained over time [27,28]. Syntheses of supervised interventions describe the frequency, intensity, and supervision parameters associated with benefit and thus offer a template for implementation across the treatment pathway [13]. Consensus guidance already supports the integration of exercise into routine oncological care [7].

This review has several limitations. As a narrative review, this work did not apply a systematic search protocol or a formal risk-of-bias assessment, and the selection of literature, although directed toward the highest-quality and most recent evidence, is inevitably less exhaustive than in a systematic review. The primary literature is heterogeneous in populations, interventions, and outcome measures; several settings, active surveillance and mechanistic research in particular, rest on small samples; and in the perioperative pelvic-floor literature the methodological quality of the underlying meta-analyses and systematic reviews has been judged critically low, which constrains the strength of any recommendation [26].

An important reference point comes from outside prostate cancer. The phase 3 CHALLENGE trial in stage III and high-risk stage II colon cancer provided the first level-1 evidence that a structured exercise programme can improve hard oncological endpoints: a three-year intervention begun after adjuvant chemotherapy significantly improved disease-free survival and yielded findings consistent with longer overall survival, with a 5-year disease-free survival of 80.3% versus 73.9% for health education alone (hazard ratio for recurrence, new primary cancer, or death, 0.72) [33].

This establishes the biological and clinical plausibility of an exercise effect on survival, but it cannot be extrapolated directly to prostate cancer, which differs in biology, natural history, and treatment. No completed phase 3 trial has yet demonstrated a survival benefit of exercise in prostate cancer, and current guidelines make no such claim [22]. CHALLENGE therefore strengthens the case for adequately powered survival trials in prostate cancer without settling it.

Adequately powered randomized trials with hard oncological endpoints are needed to establish whether the biological signals observed to date translate into effects on progression or survival. Standardization of exercise prescription and outcome reporting would allow more meaningful comparison across studies, and dedicated implementation research is required to translate efficacy demonstrated under supervision into durable, real-world practice.

5. Conclusions

Exercise acts on the physiological systems that prostate cancer treatment, and ADT in particular, disturbs. Resistance training and combined resistance-plus-aerobic programmes carry the strongest evidence for limiting the loss of lean mass, the gain in fat mass, and the cardiometabolic changes that follow androgen suppression [17,20,21]. What the field now requires is adequately powered phase 3 trials with hard oncological endpoints, principally disease-specific survival and biochemical recurrence; no such trial has yet been completed in prostate cancer [22], and the evidence that structured exercise can affect these endpoints comes from colon cancer [33]. It also requires standardized prescriptions organized into phase-specific pathways, without which efficacy demonstrated under supervision will not carry over into community practice [13,28].

Disclosure

Supplementary Materials

Not applicable.

Author Contributions

Conceptualization: A.R., W.Ś.; Methodology: A.R.; Investigation: M.C., D.A., P.J.; Writing - original draft: A.R., M.C.; Writing - review and editing: W.Ś., D.A., P.J.; Visualization: A.R., W.Ś. All authors have read and agreed to the published version of the manuscript.

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Artificial intelligence was used only to enhance language quality, correct grammar, and improve the academic English of the manuscript. All scientific interpretation, critical analysis, and conclusions are entirely the authors' own.

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